Introduction

An IT support ticket is a record created when a user reports a problem — such as a network issue, login problem, or hardware fault — to the IT service desk. In a typical ITSM workflow, the user submits this request, and an agent then classifies, prioritizes, and routes it to the right team. Tickets matter because they're the main way IT problems get tracked and resolved in any organization. However, this process only starts working after the ticket is already created. Vague descriptions cause back-and-forth delays for human agents, and they also hurt automation — machine learning and deep learning models used for ticket classification and routing need clear, well-described input, and their accuracy drops on incomplete or ambiguous text. On top of this, many tickets could be avoided entirely if the answer already existed in documentation or someone had already reported the same issue.

Aim

This project aims to catch problems in an IT support ticket *before* it gets created — by checking if the request is clear enough, if the answer already exists, and if it's a duplicate of another ticket. The goal is to reduce the number of unnecessary or incomplete tickets reaching human support staff, instead of only fixing tickets after they've already been filed.

Abstract

IT service desks currently automate ticket handling only after a ticket has already been created — classifying, routing, or resolving it. This means vague requests, self-resolvable issues, and duplicate reports all get filed as tickets first, and any correction happens afterward, wasting analyst time. Little published work addresses the stage before ticket creation, where these problems could be caught earlier.

This project proposes TRO (Ticket Readiness Orchestration), a three-agent system that intervenes before a ticket is created. It checks whether a user's request is complete enough to act on, whether the issue is already answered in existing documentation, and whether it duplicates an open ticket. Only after these checks does it classify and route the ticket, attaching a plain-language reason for the routing decision. Each check is confidence-gated: if the system is not confident, it asks the user a clarifying question or escalates to a human reviewer rather than guessing.

TRO does not introduce new machine learning techniques. Its contribution is architectural — combining clarification, knowledge-base deflection, and duplicate detection into one pipeline placed earlier in the ticket lifecycle than existing systems address.

The system will be evaluated on synthetic ticket data using metrics including classification accuracy, deflection accuracy, clarification efficiency, and duplicate-detection precision/recall, compared against a baseline without these gates. Real-world outcomes such as cost savings or resolution-time reduction are not claimed, since these require production data this project does not have.

Literature Review (1/2)

Paper	Objective	Methodology	Pros/Cons	Findings
TRIANGLE (ASE 2025)	Automate end-to-end incident triage — assigning cloud incidents to the correct team via specialized, negotiating LLM agents.	Multi-agent LLM framework: semantic distillation, multi-role agents with negotiation, automated Team Info Enrichment; tested on real production incidents (20TB/day, p.6) + MSR Bug Dataset.	Pro: 97% triage accuracy, 91% engagement-time cut at production scale. Con: “missing … historical signals … essential for associating incidents with the correct team” (p.9).	Multi-agent LLMs beat single-model/rule-based triage at real scale — but only after an incident already exists.
TickIt (FSE 2025)	Automate online ticket-escalation decisions — when, to whom, and whether it duplicates an existing ticket.	LLM chain-of-thought classification, category-guided fine-tuning; embedding dedup (θ=0.88); deployed on Volcano Engine; 20,066 tickets / 654,901 messages.	Pro: 39% real MTTR reduction; live feedback→fine-tuning loop. Con: “personalized expressions from … customers can influence the performance of the LLM” (p.10).	LLM escalation + dedup cuts MTTR in production, but accuracy still depends on customer phrasing — after creation.
ClarifyMT-Bench (arXiv 2025)	Benchmark whether/when LLMs should ask clarifying questions; propose ClarifyAgent to fix under-clarification.	5-dim ambiguity taxonomy; 6 personas (incl. Refusal); 6,120 dialogues; 10 LLMs evaluated; modular ClarifyAgent (Perceiver, Forecaster, Tracker, Planner).	Pro: lifts weak models sharply (Qwen-2.5-7B 66.4% →88.0%). Con: “ClarifyAgent requires five forward passes … (once per module)” (p.8).	Tested LLMs show a consistent under-clarification bias — motivating a hard turn-cap in any deployed clarifier.
Zangari et al. (ESWA 2023)	Establish a task taxonomy for Ticket Automation; benchmark BERT/XLNet hierarchical classifiers on real ticket/bug data.	4-task taxonomy (TC, EF, TR, RE; Table 1, p.5); 3 hierarchical BERT architectures; 2 real datasets; stratified cross-validation.	Pro: rigorous real-data benchmark; embedding choice alone shifts F1 by >28%. Con: “companies may not know the set of target labels” (p.17).	Strong transformers plateau at modest macro-F1 (≈0.18–0.60, pp.13-14) on real ticket text — TRO's Metric 1 ceiling.
MAS2S (Feng et al.)	Address users' inability to fully articulate needs in task-oriented dialogue by modeling when/what to ask.	MAS2S (Multi-Attention Seq2Seq); new ~100k-dialogue ClariT dataset; benchmarked vs. UBAR on success rate, question count, oracle-distance.	Pro: question count (2.97) close to oracle-optimal (2.36) vs. UBAR (p.8). Con: “this work is just one of the first steps” (p.9).	Narrows the gap to the oracle-optimal number of clarifying questions, but the field remains at an early stage.

Literature Review (2/2)

Paper	Objective	Methodology	Pros/Cons	Findings
AIM Survey (Yu et al. 2024)	Comprehensively survey 89 papers (2008–2022) on AI-driven alert/incident management; propose a unified taxonomy.	Systematic survey (snowball + keyword search); unifies architectures into one monitoring→alert→incident pipeline; no original dataset/model.	Pro: most comprehensive AIM survey to date. Con: "benchmark datasets and uniform evaluation … missing in the literature" (p.16).	"a reported incident is recorded as a ticket" (p.3) — all 89 papers act only after that recording happens.
Fuchs et al. (TUM/SAP 2026)	Compare human vs. automated (Active Learning + Weak Supervision) annotation; examine ticket ambiguity's impact.	Case study on real SAP data; crowdsourced human labels vs. algorithmic Learning Functions needing only 10% human-labeled seed data.	Pro: 92% agreement with human labels at a fraction of the cost. Con: proposes, then declines, a pre-check: "a whole project in itself" (p.9).	Authors describe a pre-distribution ambiguity check, then decline to build it — "a whole project in itself" (p.9).
Salah et al. (Inf. Fusion 2019)	Improve incident-resolution efficiency by fusing human ticket text with machine alert data on a real telecom network.	Rule-based time-alignment correlation (time thresholds + keyword similarity); validated on real alert/ticket data, production telecom network.	Pro: validated on real data; measurably raises alert-reduction rate. Con: criteria "depend on … keywords that may depend on the considered network" (p.14).	Fusing tickets with alerts improves correlation on real data, but is rule-dependent and runs only on tickets already created.
Zicari et al. (ESWA 2022)	Build an accurate, explainable ticket classifier combining deep ensembles with LIME-style local explanation.	Deep ensemble of CNN/FFNN base classifiers; LIME-based local explanation; evaluated on real (non-synthetic) ticket data.	Pro: validated on real data; generates genuinely human-facing rationale. Con: "plan to … extend … with … Deep Active Learning" (p.15) — no feedback loop yet.	Deep ensembles + explanation improve robustness and trust in routing — explainable rationale is already a mature area.
Dhole (arXiv 2020)	Generate discriminative clarifying questions between two competing intents, without annotated clarification data.	Rule-based hybrid: question-generation model + sentence-similarity model (template + generative); human Likert ratings and coverage metrics.	Pro: needs no clarification-labeled training data. Con: "coverage of the discriminative similarity approach is still low" (p.6).	Can disambiguate two intents without labeled data, but coverage stays low — evidence against a rule-based question bank.
Sravani et al. (IJERST 2026)	Predict ticket priority, resolution status, and satisfaction using XLNet embeddings + Histogram-Based Gradient Boosting.	XLNet feature extraction → HGB multi-target classifier; reports 96.26% (satisfaction) / 95.41% (resolution); no stated dataset source.	Pro: direct precedent for TRO's planned Agent-5 upgrade. Con: no self-reported limitation in the text; unusually high accuracy is a red flag (inference, not the paper's claim).	Very high accuracy on leakage-prone targets with no discussion of provenance or train/test method — treat with caution.

Research Gaps

Research Gap 1: No system intervenes before a ticket is created

- All reviewed automation — triage, escalation, classification — acts only after a ticket record already exists. No reviewed system decides whether a ticket should be created at all.

Research Gap 2: LLMs consistently under-ask clarifying questions

- Language models tend to proceed with incomplete information instead of asking for clarification, even when tested specifically on this behavior.

Research Gap 3: Knowledge-base retrieval is used as a passive aid, not a gate

- Existing systems use KB/documentation retrieval to help a human or generate an answer, but nothing decides whether retrieval is confident enough to prevent ticket creation altogether.

Research Gap 4: Duplicate/incident correlation happens after tickets are filed, not before

- Correlation and deduplication techniques exist, but they operate on tickets or alerts that have already been created, not on incoming messages before ticket creation.

Objectives

- **Design** a confidence-gated, multi-agent architecture that intervenes before ticket creation, addressing an identified gap in existing ITSM automation literature.
- **Develop** intake and clarification mechanisms that assess request completeness and ask targeted follow-up questions only when needed, capped at a fixed number of turns.
- **Implement** knowledge-base deflection and duplicate/incident correlation modules to reduce unnecessary ticket creation before dispatch.
- **Integrate** the pipeline end-to-end with Jira Service Management for ticket dispatch and PostgreSQL for audit logging of all gate decisions.
- **Evaluate** system performance on synthetic ticket data using metrics including classification F1, deflection accuracy, correlation precision/recall, and LLM confidence calibration, benchmarked against a no-gate baseline.

SDG Alignment:

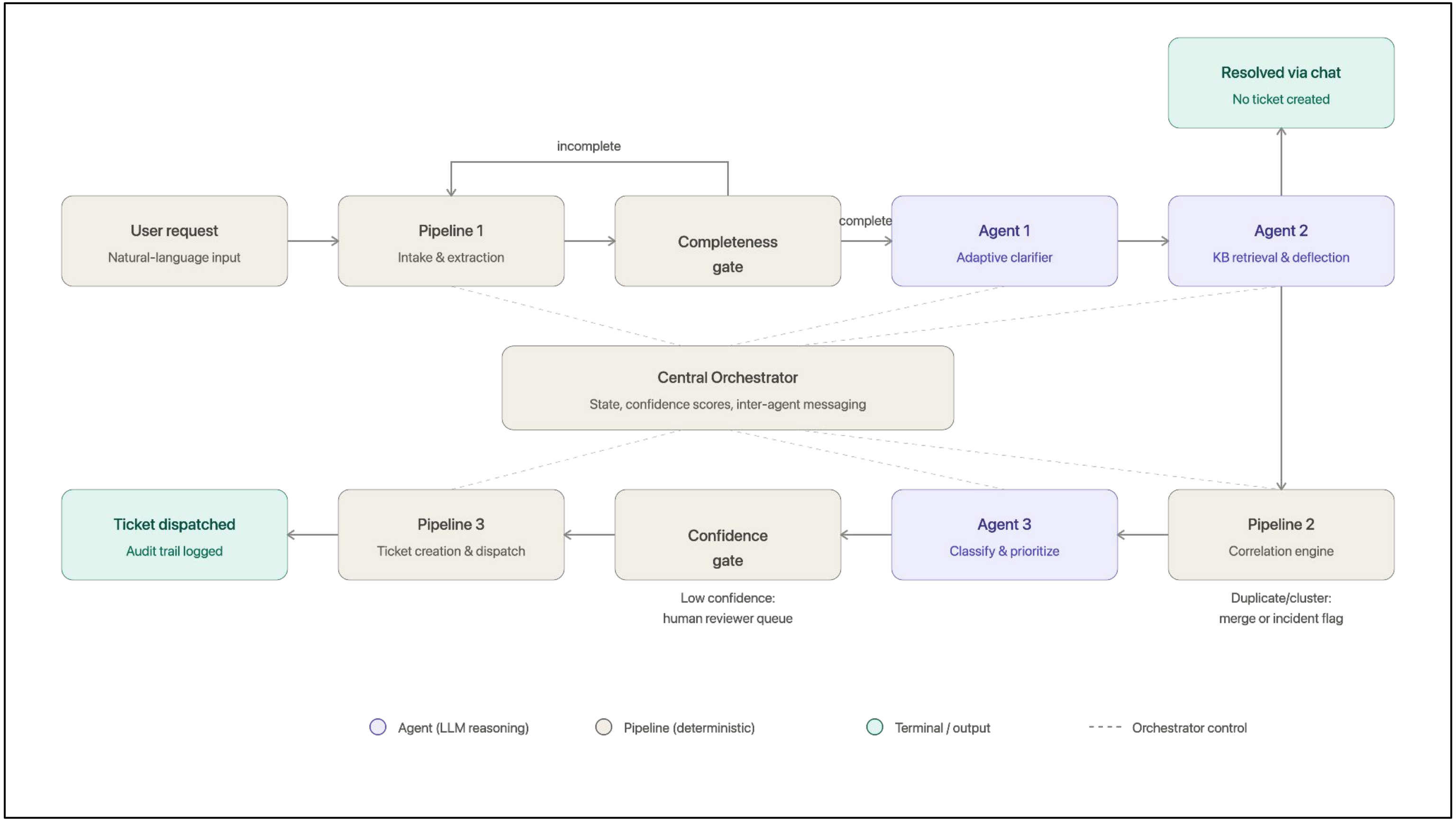
SDG 9 – Industry, Innovation and Infrastructure: TRO proposes a novel multi-agent architecture that repositions ticket automation earlier in the IT service lifecycle, contributing an innovative systems-level design to enterprise IT infrastructure tooling.

TRL:

TRL 3 – Experimental Proof of Concept: Core mechanisms — confidence-gated clarification, KB deflection, and duplicate correlation —validated at small scale, demonstrating initial technical feasibility ahead of full pipeline integration.

Project Outcome: Submission of a research paper to a Scopus-indexed or IEEE conference, presenting the pre-submission gated multi-agent architecture and its evaluation against a no-gate baseline.

Architecture



References (1/2)

[1] Z. Yu *et al.*, "Triangle: Empowering incident triage with multi-agent," in *Proc. 40th IEEE/ACM Int. Conf. Automated Software Engineering (ASE)*, 2025.

[2] F. Liu, X. He, T. Zhang, J. Chen, Y. Li, and L. Yi, "TickIt: Leveraging large language models for automated ticket escalation," in *Proc. ACM Int. Conf. Foundations of Software Engineering (FSE)*, Trondheim, Norway, 2025.

[3] S. Luo, Y. Huang, M. Li, S. Meng, F. Liu, Z. Hu, J. Feng, and Q. Liu, "ClarifyMT-Bench: Benchmarking and improving multi-turn clarification for conversational large language models," *arXiv:2512.21120*, 2025.

[4] A. Zangari, M. Marcuzzo, M. Schiavinato, A. Gasparetto, and A. Albarelli, "Ticket automation: An insight into current research with applications to multi-level classification scenarios," *Expert Systems with Applications*, vol. 225, art. 119984, 2023.

[5] Y. Feng, H. A. Rahmani, A. Lipani, and E. Yilmaz, "Towards asking clarification questions for information seeking on task-oriented dialogues," *arXiv:2305.13690*, 2023.

[6] Q. Yu, N. Zhao, M. Li, Z. Li, H. Wang, W. Zhang, K. Sui, and D. Pei, "A survey on intelligent management of alerts and incidents in IT services," *Journal of Network and Computer Applications*, vol. 224, art. 103842, 2024.

[7] S. Fuchs, J. Schnellbach, H. Wittges, and H. Krcmar, "Human vs. automated data annotation: Labeling the data set for an ML-driven support ticket classifier," *Data & Knowledge Engineering*, vol. 162, art. 102534, 2026.

[8] S. Salah, G. Maciá-Fernández, and J. E. Díaz-Verdejo, "Fusing information from tickets and alerts to improve the incident resolution process," *Information Fusion*, vol. 45, pp. 38–52, 2019.

References (2/2)

[9] P. Zicari, G. Folino, M. Guarascio, and L. Pontieri, "Combining deep ensemble learning and explanation for intelligent ticket management," *Expert Systems with Applications*, vol. 206, art. 117815, 2022.

[10] K. D. Dhole, "Resolving intent ambiguities by retrieving discriminative clarifying questions," *arXiv:2008.07559*, 2020.

[11] V. Sravani, D. Thalla, V. Anusree, M. Mubin, C. Eshwar, and P. Meghana, "Scalable customer support ticket analysis using hybrid NLP and ensemble-based classification models," *Int. J. Engineering Research and Science & Technology*, vol. 22, no. 2(1), pp. 459–471, 2026.

[12] F. Al-Hawari and H. Barham, "A machine learning based help desk system for IT service management," *Journal of King Saud University – Computer and Information Sciences*, vol. 33, no. 6, pp. 702–718, 2021.

[13] J. Schad, R. Sambasivan, and C. Woodward, "Predicting help desk ticket reassignments with graph convolutional networks," *Machine Learning with Applications*, vol. 7, art. 100237, 2022.

[14] M. Sohail, M. Shukla, and A. Gupta, "AI-powered complaint management and ticketing system," *Int. Research J. Modernization in Engineering Technology and Science*, vol. 7, no. 12, pp. 901–907, 2025.

[15] Prathik B. K., Manovikyath L., Likhith G. Gowda, Yashas Kumar S., and S. Manoli, "Online chatbot's ticketing system," *Int. J. Innovative Research in Science, Engineering and Technology*, vol. 14, no. 4, pp. 9865–9873, 2025.

[16] A. Taha, D. Mabrouk, A. Tamer, M. Tarek, F. Alaa, M. El-Fateh, and F. Romany, "Smart Serve: Redefining customer support with AI-driven ticketing intelligence," *Advanced Sciences and Technology Journal*, vol. 3, no. 1, art. P1069, 2026.